

Alp Berke Ardic

Chicago, IL | alpberke.ardic@gmail.com | alpbardic.github.io
linkedin.com/in/alpberkeardic | github.com/alpbardic | Google Scholar

SUMMARY

PhD candidate at UIC working on decentralized LLM fine-tuning up to 12B parameters, distributed training algorithms, and model-parallel inference on edge devices. First author on 7 of 8 papers (2 published, 2 accepted at workshops, 4 under review).

EDUCATION

University of Illinois Chicago, PhD in Electrical and Computer Engineering, GPA 4.0/4.0 Aug 2022 – Present
Advised by Prof. Erdem Koyuncu. Teaching Assistant for Machine Learning, Deep Learning, and Pattern Recognition (100+ students).
University of Illinois Chicago, MS in Electrical and Computer Engineering, GPA 4.0/4.0 Aug 2022 – Apr 2026
Bilkent University, BS in Electrical and Electronics Engineering, Ankara, Turkey 2017 – 2022

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Assistant, University of Illinois Chicago, Chicago, IL Sep 2022 – Present

- Developed **RUMOR** for **decentralized LLM fine-tuning** when nodes with rising loss train less often and their data becomes under-represented. It reweights updates using only local signals (ICLR 2027, under review, extends an accepted NeurIPS 2026 workshop paper).
- Built it in PyTorch with Hugging Face Transformers and PEFT, running rank-8 LoRA on 7 pretrained language models up to 12B parameters (GPT-2, Pythia, Qwen2.5, Mistral-Nemo) in a simulated 8-node network on an NVIDIA GPU. At 12B under strong feedback it had 0 of 10 unstable runs vs. 3 of 10 for windowed inverse-probability weighting, and 1 vs. 27 over 120 small-transformer feedback runs.
- Designed **Random Walking Snakes** for decentralized training, splitting a model across edge-device groups to train about 40% faster than random-walk learning in simulation, then extended it to non-IID DistilBERT training (IEEE LANMAN 2023, IEEE TMC under review).
- Created coordinator-free **model-parallel inference** that assigns DNN layer blocks to edge devices by decentralized graph coloring, cutting assignment conflicts by 92–99% and hop-based latency by 10–11.8% in simulation (IEEE MILCOM 2025, IEEE TPDS under review).

Research Intern, TUBITAK SAGE (national defense R&D institute), Ankara, Turkey Sep 2021 – May 2022

- Raised real-time YOLO object detection accuracy by 18% under motion blur using custom augmentation and hyperparameter tuning.
- Sped up the inference preprocessing pipeline by 25% for real-time, high-speed imaging.

Robotics Researcher, Bilkent Miniature Robotics Lab | inspection UAVs, reverse-engineered commercial drones Jan 2021 – Nov 2021

Digital Design Engineering Intern, Koc Information and Defense Technologies | SDR interfaces on FPGAs in VHDL Aug 2021 – Sep 2021

Embedded and Camera Systems Intern, ASELSAN | camera interface built from scratch in C on a microcontroller Jun 2021 – Aug 2021

AI and Computer Vision Intern, TWIN Science & Robotics | real-time CNN vision at about 15 FPS on Raspberry Pi Jul 2019 – Sep 2019

SELECTED PROJECTS

On-Device CNN Compression for ESP32 | ECE 594 team project (4 members), UIC Oct 2025 – Dec 2025

- Chose the model and its compression (structured pruning, adaptive pooling, INT8 post-training quantization) and wrote the Python training code, shrinking a CIFAR-10 CNN 42× (2.2 MB to 52 KB) to fit an ESP32 with 312 KB of DRAM at a 3.4-point accuracy cost.

Reinforcement Learning MAC Protocol in ns-3 | ECE 533 solo project, UIC Oct 2025 – Dec 2025

- Implemented RL-MAC in ns-3 (C++) and a Python Q-learning agent that earned 1.5–2× the reward of random and fixed-slot baselines.

PUBLICATIONS

- A. B. Ardic**, E. Koyuncu, “Who Gets Heard by Gossip? Correcting Participation Bias in Decentralized Language-Model Fine-Tuning,” *NeurIPS 2026 Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM)*, accepted.
- A. B. Ardic**, E. Koyuncu, “When Does Spatial Cooperation Help? Contamination Regimes in Decentralized Online Spectrum Anomaly Learning,” *IEEE MILCOM 2026 Workshop*, accepted.
- A. B. Ardic**, M. Dwyer, E. Koyuncu, “Decentralized Model-Distributed Inference,” *IEEE MILCOM 2025*.
- A. B. Ardic**, H. Seferoglu, S. El Rouayheb, E. Koyuncu, “Random Walking Snakes for Decentralized Learning at Edge Networks,” *IEEE LANMAN 2023*.
- A. B. Ardic**, E. Koyuncu, “RUMOR Has It: Correcting Participation Bias in Decentralized Language Model Fine-Tuning,” *ICLR 2027*, under review.
- R. Dowlatshahi, ..., **A. B. Ardic**, ..., E. Koyuncu (18 authors), “HotSight: A Paired RGB-Thermal UAV Dataset and Benchmark for Wildfire Combustion-State Segmentation and Hotspot Localization,” *WACV 2027*, under review.
- A. B. Ardic**, M. Dwyer, E. Koyuncu, “Cost-Aware Asynchronous Persistent Graph Coloring for Decentralized Model-Distributed Inference,” *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, under review.
- A. B. Ardic**, H. Seferoglu, S. El Rouayheb, E. Koyuncu, “Random Walking Snakes for Decentralized Learning of Heterogeneous Data at Edge Networks,” *IEEE Transactions on Mobile Computing (TMC)*, under review.

TECHNICAL SKILLS

Machine Learning | large language models (LLMs), parameter-efficient fine-tuning (LoRA), distributed training algorithms, decentralized optimization, federated learning, model-parallel inference, model compression, computer vision, reinforcement learning

Frameworks | PyTorch, Hugging Face Transformers, PEFT, TensorFlow, OpenCV

Languages and Tools | Python, C/C++, MATLAB, VHDL, Linux, ns-3, FPGA, Raspberry Pi, Arduino